

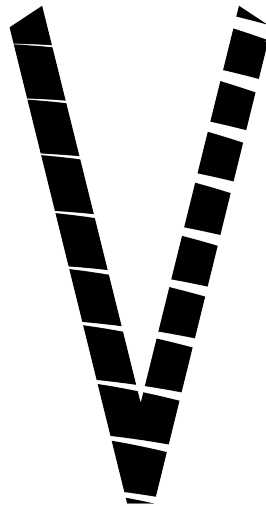
STMicroelectronics LSM6DSV320XTR Breakout Board

Project Manager

Ben Teigland

Goal of Project:

Design and spin a PCB that allows people to interface with the LSM6DSV320XTR dual accelerometer architecture. The goal will be to provide a breakout board in the MPU-6050 form factor for use as a switch out to gain usage of a higher performing accelerometer in identical circuits. This is driven by the needs of the Bethel Rocket Club needing a new IMU.



Vector Devices

Table of Contents

Project Outline.....	3
Items to be Included.....	3
Technical Specifications.....	3
Power.....	3
Communication Protocols.....	3
Footprint Goals.....	4
Mechanical Dimensions.....	5
Connections to be Made.....	5
Required Components.....	7
Preliminary Bill of Materials.....	7
Deliverables.....	8
Deliverable 1: KiCad Schematic (.kicad_sch).....	8
Deliverable 2: KiCad PCB Layout (.kicad_pcb).....	8
Deliverable 3: Gerber Files (.zip).....	8
Deliverable 4: Bill of Materials BOM (.csv).....	8
List of Deliverables.....	9
Timeline.....	9
Workspace Breakdown.....	9
ClickUp.....	9
GitHub.....	9
KiCad 9.0.....	9

Project Outline

STMicroelectronics has released a new IMU that adds the capability of dual accelerometer. This includes a $\pm 16G$ accelerometer and a $\pm 320G$ while also including the ± 4000 dps gyroscope. A board like this can be used in rocketry to better predict 3D orientation as the lower Gs can be used for fine movements and the higher Gs can be used for general flight. In general, the main goal is to get it into the hands of the open source community and the Bethel Rocket Club in order to get the new sensor some tread and see what programmers can do with it.

Items to be Included

The board will include a LSM6DSV320XTR sensor with the required supporting components as per the datasheet. The board will include equally space holes (2.54mm) that we will be able to solder headers onto to connect to typical breadboard sizing. It will also include an LDO for power optimization along with a LED indicator for power.

Technical Specifications

The printed circuit board will require a few specific implementations. These are compiled with the goal of helping the design process.

Power

The board will have a VDD power input. This will power the LSM6DSV320XTR sensor and create a high line for the pull up resistors. No VDDIO functionality will be added from the user accessible pinouts. The board itself will be powered by a 3.3V input. Since no access to VDDIO is provided, users will not be able to power the board with a 1.8V supply with a 3.3V logic input. However, a LDO will be mounted on the board to power the LSM6DSV320X at 1.8V. This will limit low power mode functionality as the pull up resistors of 10K will consume about 2 mW of current in low power mode which is only half of the estimated 5.5 mW the board will run at (not including the LED). This assumes a low quiescent current and a 50 uA input on VDDIO based on typical I/O activity of I2C communications. Ultimately, this power scheme is intended to provide maximal use at high performance mode and will limit its use at low power modes significantly.

Communication Protocols

The system will allow users to utilize the I2C, I3C, and SPI communication protocols. However, due to the existence of pull ups on the SDA and SCL data lines, it should be expected that I3C will not be able to run at its advertised rate of 12.5 MHz. All communications lines must be 3.3V since VDDIO

will be powered to 3.3V. I2C has been prioritized in this design as it is most familiar and accessible by most microcontrollers. It will be labeled for a typical I2C setup as this is an easy and common way to interface with these types of sensors. According to the datasheet, the pinouts will provide provisions for the use of Mode 2 that is shown in Figure 1.

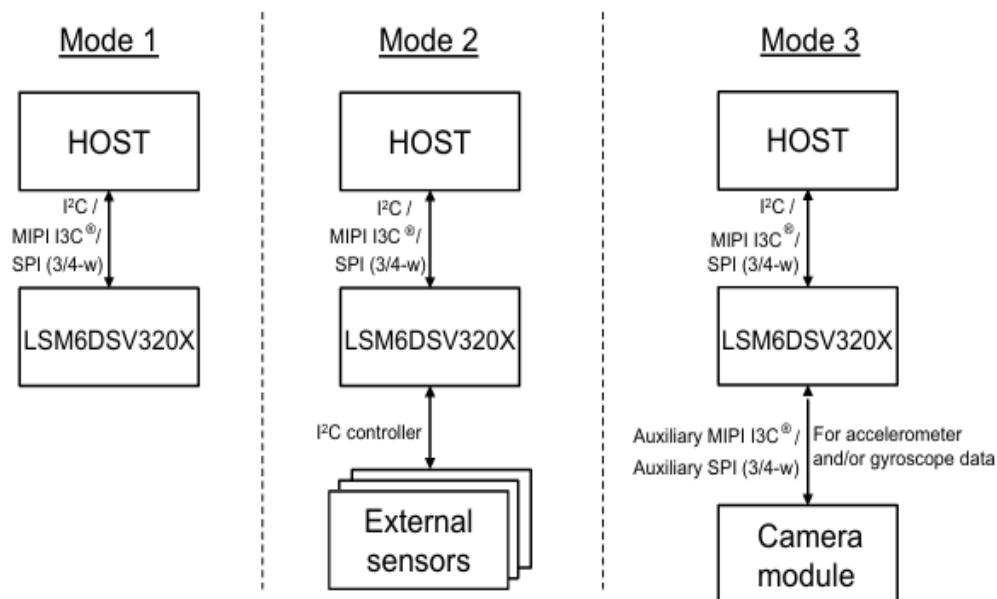


Figure 1: Description of modes defined on the LSM6DSV320XTR datasheet by STMicroelectronics (DS14623, Rev 1, Pg 10)

The intended mode for this project is Mode 2, which includes Mode 1 functionality. This allows the LSM6DSV320 to interact with and daisy chain with other I2C sensors. It also provides a CS pin so that users of the board can switch between I2C and SPI. Mode 3 has been omitted since the intended use of the board is not for camera applications as it would be beneficial to directly integrate this sensor into the camera. This removes the need for SDO_aux and OCS_aux leaving pin 11 and 10 as NC.

Footprint Goals

The goal of this design is to make this board a perfect replacement for the MPU-6050 breakout boards. This is due to the fact the Bethel Rocket Club uses this board exclusively right now and this footprint is one I am comfortable with. The features of this include 8 Pin headers. However, this will severely limit the functionality of the board. Instead a 10 Pin configuration will be used to gain full use of Mode 2.

Mechanical Dimensions

The board will be loosely based on the MPU-6050 breakout board. The parts that will align are the 8 pins spaced 2.54mm apart and the mounting holes placed in the same spot in relation to this. The dimensions of the MPU-6050 can be seen in Figure 2. The breakout board will be extended to a total of 26mm. This will shift the solder pad to wall dimensions (going from the left to circle) from 1.36mm to 1.57mm (with the addition of the two pins). The mounting hole to wall dimensions will adjust from 2.5mm to 5.25mm. The rest of the dimensions are to remain the same.

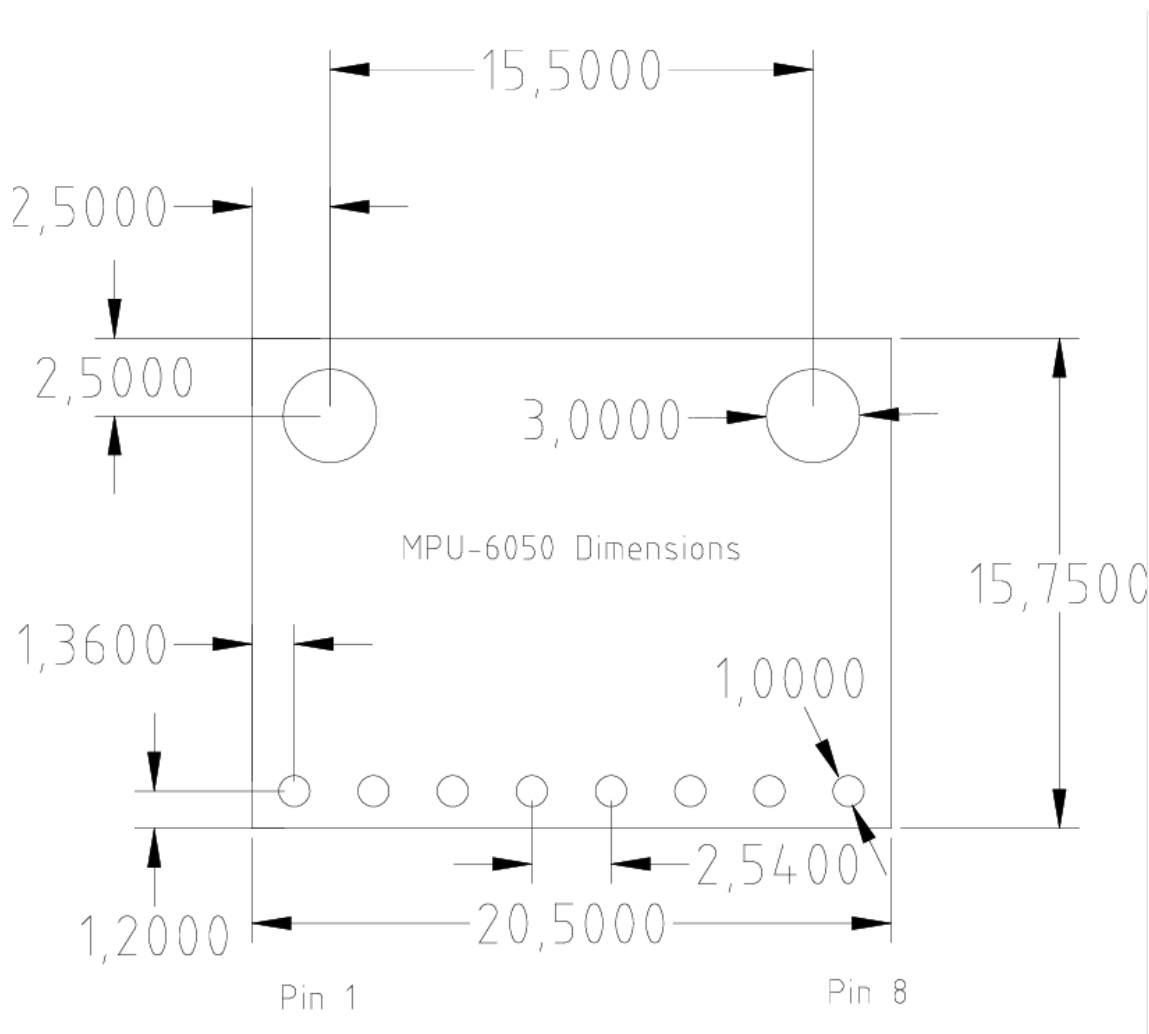


Figure 2: Mechanical Dimensions of the MPU-6050.

Connections to be Made

The breakout board will mirror the first 8 pins of the MPU-6050 to act as a solid replacement instead of a perfect one. Figure 2 show sthe pin orientation number 1-8 from left to right. This assumes the IMU device is on the top side and VCC is labeled pin 1. For the purposes of the new

board, this mapping will begin at Pin 2 since we will be adding a pin to the outer edges of the 8 pin setup to create a 10 pin board.

NC → CS: 1

VCC → VDDIO and LDO Input: 2

GND → GND: 3

SCL → SCL: 4

SDA → SDA: 5

XDA → CSDA: 6

XCL → CSCL: 7

ADO → SDO: 8

INT → INT1: 9

NC → INT2: 10

A 10 Pin footprint was chosen so that the board could fully utilize the INT1, INT2, and CS functionality. CS adds the ability to run the board in both I2C, I3C, and SPI mode. CS should be pulled to low to enable SPI mode. The board includes a pull up resistor on CS so that the out of box mode is I2C. If SPI is desired, the CS line should be pulled low. The interrupts can be configured for certain types of applications as defined by STMicroelectronics in the datasheet for the LSM6DSV320XTR sensor.

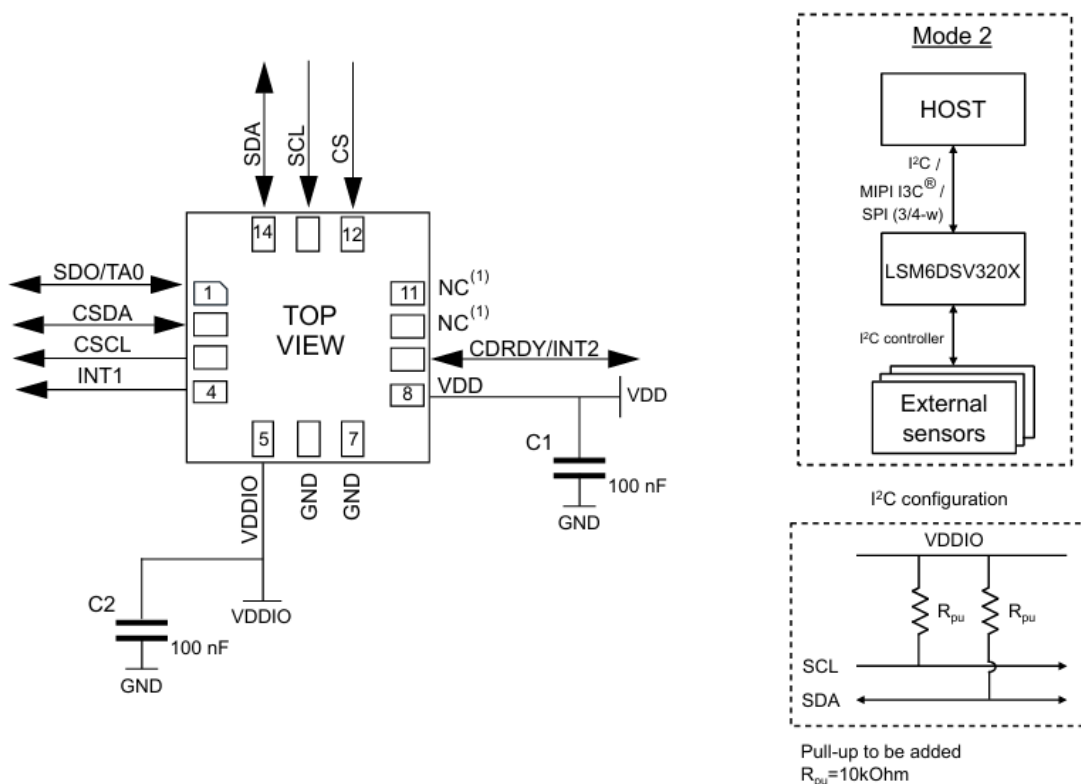


Figure 3: Wiring diagram provided for LSM6DSV320X from the datasheet by STMicroelectronics (DS14623, Rev 1, Pg 50)

Required Components

The board will require components beyond the LSM6DSV320X sensor module as referenced by the Figure 3. These extra components are listed as follows.

- 2 100 nF Ceramic Capacitors
- 3 10K Ohm Resistors for Pull Ups on the SDA , SCL, and CS lines.
- 1.8V LDO (with accompanying capacitors as chosen LDO datasheet specifics)
- 1 680 Ohm Resistor for LED
- 1 SMD LED

The components connections can be seen in Figure 3. The VDDIO line will be provided from the pin head on the breakout board labeled VCC. The VDD line will be provided from the output of the LDO. Please note the chosen LDO must meet the 0.67 mA requirement, with a preference for much higher, for the board to be operational.

Preliminary Bill of Materials

Below is the beginning of a Bill of Materials. This will adapt for each configuration depending on whether the board is being built by JLCPCB, PCBWay, or built from scratch from DigiKey parts.

Part	Description	Quantity
LSM6DSV320	IMU sensor to be interfaced with	1
100 nF Ceramic Capacitor	Filtering Capacitor for power lines	2
10K Ohm Resistor	Pull Up resistor for I2C bus data integrity and CS	3
1.8V LDO	For supplying 1.8V to VDD	1
LDO Capacitors	Specified by chosen LDO	(See LDO Datasheet)
680 Ohm Resistor	LED current limiting resistor	1
SMD LED	For power indication of light	1

These are the base requirements of the board. A much more detailed BOM will be required in order to fully complete the project.

Deliverables

This project will require some well defined deliverables. A short list of each item will be provided at the end of this section after the explanations of each step. The deliverables will be listed in each step below with explanations.

Deliverable 1: KiCad Schematic (.kicad_sch)

The first deliverable will need to be a created schematic layout in KiCad. The schematic will be a singular page with all important details stored in the proper sections. Every symbol will include a link to where the intended manufactured part will be. All **Net Classes** will be defined in this stage.

Deliverable 2: KiCad PCB Layout (.kicad_pcb)

The second deliverable will building the actual layout of the breakout board in the KiCad PCB Editor. Utilizing the connections created by the KiCad schematic and the net classes, the board will be created based off of the MPU-6050 footprint.

Deliverable 3: Gerber Files (.zip)

The third deliverable will be to create the Gerber Files for both JLCPCB and PCBWay. The files will be stored on the GitHub with clear directions to utilize it.

Deliverable 4: Bill of Materials BOM (.csv)

The fourth deliverable will be to generate a bill of materials in the format for JLCPCB, PCBWay, and DigiKey. This will allow flexibility in manufacturing for different users.

List of Deliverables

Deliverable 1: KiCad Schematic (.kicad_sch)

Deliverable 2: KiCad PCB Layout (.kicad_pcb)

Deliverable 3: Gerber Files (.zip)

Deliverable 4: Bill of Materials BOM (.csv)

Timeline

The project should be finished within 1 month of conception. This would mark the finish date as **June 27th**. This includes the entire GitHub and a board ordered from a manufacturer. A separate project will be introduced to make the library firmware for the board once the board has been assembled and shipped. This will leave the month of July for programming the board.

Workspace Breakdown

My intention is to setup the workspace that is professional and efficient. ClickUp and GitHub will be utilized to streamline the project. Ideally the work should be documented, professional, flexible, and ultimately good.

ClickUp

Task breakdown and timeline tool. The intention is to use ClickUp to stay on track with project deliverables and timelines.

[ClickUp Project Folder](#)

GitHub

The GitHub will operate as the open source database where all project files will be stored. The intent is to allow others to manufacture and change the board depending on their design desires.

[GitHub Repository](#)

KiCad 9.0

The board will be designed in KiCad 9.0. All schematics and footprints will be created and exported from KiCad. The files will be prepped for manufacturing from JLCPCB and PCBWay for full assembly from these manufacturers. It will also includes provisions needed to order parts from DigiKey for assembly.

Revision History

Revision	Comments
2	Addition of LED as well as adjusted mechanical description to match 1.36 widths instead of 2.5
1	Original Creation of Design Guide